

Notes: Gieczewski & Kosterina (RES, 2024)

Experimentation in Endogenous Organizations

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1 Big picture

- **Question.** Why do some organizations persist with failing policies even after long sequences of bad news?
- **Core idea.** Information affects both beliefs and membership. Bad news makes everyone more pessimistic, but it also makes pessimists exit the organization. This selection can leave the organization controlled by increasingly optimistic members.
- **Baseline organization.** Agents can freely enter or leave an organization. Members receive the flow payoff from the organization's current policy and vote over future policy.
- **Policies.** The organization can use a safe policy with known payoff or a risky policy whose quality is unknown. If the risky policy is good, successes arrive randomly; if it is bad, no successes arrive.
- **Main contrast.** Standard collective experimentation models with fixed membership often generate under-experimentation. This paper shows that endogenous membership can instead generate over-experimentation.
- **Main result.** Under sufficient conditions, perpetual experimentation is an equilibrium: the organization never abandons the risky policy despite accumulating negative evidence.
- **Mechanism.** The current pivotal member becomes more pessimistic over time, but the identity of the pivotal member shifts toward ex ante optimists as conservatives leave. The selection effect can dominate the learning effect.
- **Sharp implication.** Perpetual experimentation is more likely when agents are patient, the population contains enough optimists, and the outside option is attractive relative to the safe policy.
- **Extensions.** Heterogeneous outcomes can make over-experimentation stronger: optimists can ally with agents who are guaranteed winners from the risky policy.
- **Corrective force.** When control belongs to shareholders who own all future payoffs, capital gains and tradable shares can curb the selection force and rule out perpetual experimentation in the model.
- **Economic takeaway.** Exit can weaken voice. If sceptics leave rather than fight internally, a declining organization can become more radical and persist with the failing experiment.

2 Model primitives and measurement

2.1 Agents and beliefs

- Time is continuous, $t \in [0, \infty)$.
- There is a continuum of agents indexed by $x \in [0, 1]$.
- Agent x has prior belief x that the risky policy is good.
- Priors are distributed with continuous density f over $[0, 1]$.
- All agents discount at rate γ .

Interpretation: The index x is both an identity and a belief. Higher- x agents are ex ante more optimistic about the risky policy.

2.2 Policies and payoffs

The organization has a safe policy and a risky policy.

- If the safe policy is used, members receive flow payoff s .
- Non-members receive outside option flow payoff a .

- If the risky policy is good, successes arrive according to a Poisson process with rate λ .
- Each success gives members a lump-sum payoff h .
- If the risky policy is bad, no successes occur.

Let

$$g = \lambda h \quad (1)$$

be the expected flow payoff of the good risky policy. The maintained ordering is

$$0 < a < s < g. \quad (2)$$

Interpretation: The safe policy is better than the outside option, but a good risky policy is better than the safe policy. The bad risky policy is worst.

2.3 Learning after failures

If the organization experiments unsuccessfully until time t , agent x 's posterior belief is

$$p_t(x) = \frac{xe^{-\lambda t}}{xe^{-\lambda t} + 1 - x}. \quad (3)$$

After a success, all posteriors jump to 1.

Interpretation: Failure creates common negative information: every agent's posterior falls. But higher-prior agents remain more optimistic than lower-prior agents.

2.4 Membership threshold

Under experimentation, agent x wants membership if

$$p_t(x)g \geq a. \quad (4)$$

Thus the set of members is an interval

$$[y_t, 1], \quad (5)$$

where the marginal member y_t solves

$$p_t(y_t) = \frac{a}{g}. \quad (6)$$

Using Bayes' rule,

$$y_t = \frac{a}{a + (g - a)e^{-\lambda t}}. \quad (7)$$

Interpretation: As failures accumulate, y_t rises. More pessimistic agents leave, so the organization shrinks and its membership becomes more optimistic.

2.5 Voting and the pivotal member

The policy is chosen by the median member. Under experimentation without success, the pivotal member is

$$m_t = \text{median of } F \text{ restricted to } [y_t, 1]. \quad (8)$$

Interpretation: Bad news raises y_t , and this can raise m_t . Thus the median member can become more optimistic in prior type even though every individual's posterior is declining.

3 Equilibrium and main propositions

3.1 Stopping times

Let $T \subseteq [0, \infty)$ be the set of stopping times at which the organization switches to the safe policy if no success has occurred. Let

$$t_0 = \inf T \quad (9)$$

be the equilibrium stopping time. If $T = \emptyset$, write $t_0 = \infty$.

Interpretation: The model is reduced to a sequence of policy choices by pivotal members along the no-success path.

3.2 Continuation values

Let $V_T(y)$ be the continuation payoff of an agent with current belief y who expects experimentation to continue for length T . Let

$$V(y) = \lim_{T \rightarrow \infty} V_T(y) \quad (10)$$

be the value of perpetual experimentation. The payoff from stopping and switching to the safe policy is

$$\frac{s}{\gamma}. \quad (11)$$

Interpretation: The pivotal member compares the continuation value of experimentation with the discounted payoff from the safe policy.

3.3 Proposition 1: perpetual experimentation

Perpetual experimentation is an equilibrium if and only if

$$V(p_t(m_t)) \geq \frac{s}{\gamma} \quad \text{for all } t. \quad (12)$$

It is the unique equilibrium if the inequality is strict for all t .

Interpretation: The condition says that every future pivotal member must prefer never-ending experimentation to stopping. Because the pivotal identity changes over time, this can hold even though every fixed agent eventually becomes pessimistic.

3.4 Why perpetual experimentation can be excessive

Perpetual experimentation can be excessive from the perspective of almost all agents. A fixed agent with prior below 1 eventually becomes arbitrarily pessimistic along a no-success path. But the organization's membership continually selects toward agents with higher priors.

Interpretation: The organization can persist not because all agents remain optimistic, but because the pessimists have exited and lost voting rights.

3.5 Proposition 2: sufficient conditions and bounds

The paper computes lower bounds or closed-form expressions for

$$\inf_{t \geq 0} V(p_t(m_t)). \quad (13)$$

For non-decreasing prior densities, the infimum of the pivotal member's posterior is

$$\inf_{t \geq 0} p_t(m_t) = \frac{2a}{g+a}. \quad (14)$$

For arbitrary full-support densities, the general bound is based on

$$p_t(m_t) \geq p_t(y_t) = \frac{a}{g}. \quad (15)$$

Interpretation: The prior distribution matters through how optimistic the median member of the surviving organization remains after repeated failure.

3.6 Corollary 1: outside option and selection

If the outside option a is close enough to the safe payoff s , perpetual experimentation obtains.

Interpretation: When a is close to s , sceptics have little reason to stay and vote for the safe policy. Exit becomes tempting, so the organization becomes dominated by optimists.

3.7 Proposition 3: comparative statics

Perpetual experimentation is easier to sustain when

- agents are more patient, i.e. γ is lower;
- learning is faster, i.e. λ is higher holding $g = \lambda h$ fixed;
- the safe payoff s is lower;
- the outside option a is higher;
- the prior distribution shifts upward in the monotone likelihood ratio order.

Interpretation: Each comparative static either makes experimentation more attractive or makes selection stronger by pushing pessimists out more quickly.

3.8 Proposition 4: finite experimentation

When perpetual experimentation is not an equilibrium, the model can have equilibria with finite stopping times. The initial pivotal member can experience too much or too little experimentation relative to the policy she would choose if she controlled the organization forever.

Interpretation: The organization is not mechanically biased toward over-experimentation in every parameter region. The selection effect and the loss-of-control effect work in opposite directions.

4 Extensions

4.1 Public news

With public news at time 0, the organization can experiment more after bad news than after good news.

Interpretation: Bad public news can drive out sceptics and radicalize the organization, while good news can attract marginal sceptics and make the median more conservative.

4.2 Heterogeneous outcomes

The paper extends the model so that some agents receive different payoffs from successful experimentation. Optimists can join forces with ex post guaranteed winners.

Interpretation: Heterogeneous payoffs strengthen the over-experimentation force because support for experimentation can come from both belief optimism and private benefits.

4.3 Continuous membership and tradable shares

The paper also studies a model of a firm controlled by shareholders. Agents can choose continuous ownership stakes, trade shares, and vote in proportion to shares. With risk-averse shareholders and fixed firm size, perpetual experimentation is impossible.

Interpretation: Tradable shares create a corrective force: optimists who want control must buy risky claims to all future payoffs, including capital gains and losses. Ownership internalizes more of the future consequences of experimentation.

4.4 Proposition 7: shareholder control

The shareholder-control extension implies:

1. there is no equilibrium with perpetual experimentation;
2. share demand $q_t(x)$ is increasing in optimism x ;
3. under logarithmic preferences, share demands become more selected toward optimists over time.

Interpretation: Selection still exists under shareholder control, but it is not strong enough to support endless experimentation because voting power requires ownership of risky future payoffs.

5 Identification and empirical design implied by the theory

5.1 What the model would predict empirically

- Organizations with easy exit should be more prone to persistence in failed experiments.
- After bad news, membership should become more optimistic or ideologically extreme.
- Policies supported by remaining members may become more radical after failures.
- Organizations with owner-shareholder control should be less prone to endless experimentation than member-run organizations with free exit.

Interpretation: The model turns membership turnover into a key observable. Persistence can reflect who remains inside the organization, not simply what all original members believe.

5.2 Empirical settings suggested by the paper

- Political parties and social movements after electoral or policy failures.
- Startups whose less committed employees leave after failed milestones.
- Cooperatives where pessimistic members exit instead of voting for conservative policy.
- Reform movements in which sceptics emigrate or opt out, leaving a more pro-reform coalition.

Interpretation: The best empirical tests would track both policy persistence and endogenous membership composition over time.

5.3 Limits of the theory

- The baseline model abstracts from strategic bargaining, leadership, and communication.
- Learning is stylized as Poisson successes and no-success failures.
- The median-voter formulation simplifies internal organizational politics.
- The model identifies a mechanism rather than estimating its empirical magnitude.

Interpretation: The model is valuable because it isolates a clean channel: exit can remove dissent and make collective experimentation excessive.

6 Tables and figures

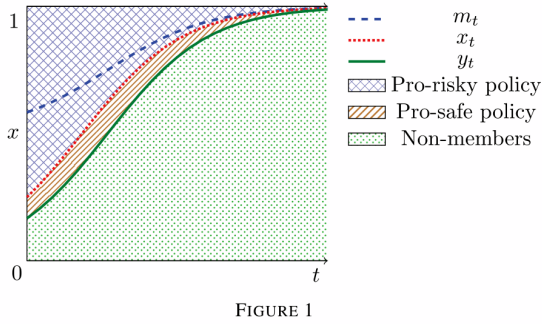
The paper contains three main figures. They are directly cropped from the paper and placed compactly below.

6.1 Figures 1 and 2: equilibrium dynamics under perpetual experimentation

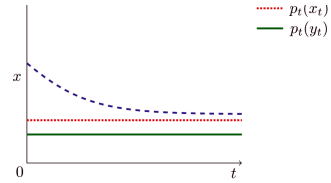
Read:

- Figure 1 plots the marginal member y_t , the indifferent agent x_t , and the median member m_t .
- As failures accumulate, y_t rises: pessimistic members leave.
- The identity of the median member m_t shifts upward, toward ex ante optimists.
- Figure 2 shows the same mechanism in posterior-belief space. The posterior of the median member does not converge to zero.

Economic message: Negative information reduces every individual's belief, but endogenous exit selects the organization toward more optimistic members. This is the central mechanism behind perpetual experimentation.



(a) Figure 1: pivotal, indifferent, and marginal agents.



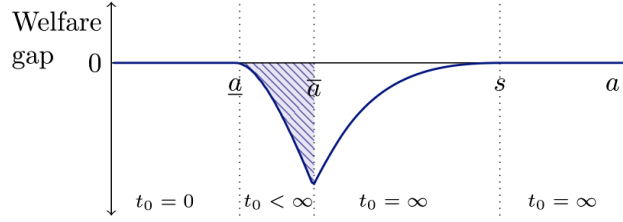
(b) Figure 2: posterior beliefs on the equilibrium path.

6.2 Figure 3: welfare gap for the initial pivotal member

Read:

- Figure 3 plots the welfare loss of the initial pivotal member relative to her first-best stopping time.
- For high a , perpetual experimentation can be optimal because the outside option is close to the safe policy.
- For intermediate a , multiple equilibria can arise and welfare losses can be largest.
- For low a , the initial pivotal member can stop close to her preferred time.

Economic message: The same exit force that sustains experimentation can produce welfare losses for the original pivotal member, especially when the outside option makes exit attractive but not entirely harmless.



7 Short summary

- The paper studies experimentation in organizations with endogenous membership.
- Agents differ in prior beliefs and can enter or leave the organization freely.
- Under experimentation, failures make all agents more pessimistic.
- Failures also induce pessimists to exit, making remaining members more optimistic.
- The median member may remain optimistic enough to continue experimentation forever.
- Perpetual experimentation can be excessive from almost every agent's point of view.
- Perpetual experimentation is more likely when agents are patient, the outside option is attractive, and the population contains enough optimists.
- Public bad news can paradoxically increase experimentation by driving out sceptics.
- Heterogeneous payoffs strengthen the mechanism by allowing optimists to ally with guaranteed winners.
- Shareholder control can correct the problem because voting power requires ownership of future payoffs.

8 Conclusion

8.1 Core conclusion

Gieczewski and Kosterina show that organizations can over-experiment because membership is endogenous. When bad news arrives, sceptics may leave rather than stay and vote for the safe policy. Their exit shifts control toward optimists, allowing a failing experiment to continue indefinitely.

8.2 Economic intuition

The intuition is a conflict between learning and selection. Learning pushes everyone toward pessimism, but selection removes pessimists from the voting body. If the second force dominates, the organization becomes more committed to the risky policy exactly when the evidence becomes worse.

8.3 Takeaway

The paper formalizes the idea that exit can undermine voice. Organizations do not simply aggregate the preferences of a fixed population. They also shape the population that remains eligible to decide. This makes endogenous membership a powerful force in collective experimentation and organizational rigidity.